

ECO7101 CHAPTER-21 PRODUCTION AND COST

A business Firm is an entity that employs factors of production (resources) to produce goods and services to be sold to consumers, other firms, or the government.

TWO SIDES TO EVERY BUSINESS FIRM

There are two sides to every market: a buying side (demand) and a selling side (supply). Similarly, there are two sides to every business firm: a revenue side and a cost side. We can see both sides of a firm by focusing on profit. The firm's objective is to maximize profit—the difference between total revenue and total cost:

Profit = Total revenue – Total cost

Looking at this profit equation, we can easily understand what any firm would like: its total revenue to be as high as possible and its total cost to be as low as possible. That way, its profit is as high as possible.

Total revenue is equal to the price of a good multiplied by the quantity of the good sold ($TR = P \times Q$). For example, if a business firm sells 100 units of X at \$10 per unit, its total revenue is $(100 \times \$10) = \$1,000$.

Cost is related to the production of the firm. Produce nothing, incur no costs; produce something, incur costs.

An explicit cost is a cost that is incurred when an actual (monetary) payment is made. Suppose you spent tk. 2000 on books, tk. 200 for lunch, tk. 100 for pen and pencil. All these are explicit or accounting cost. Total explicit cost = $2000 + 200 + 100 = \text{tk.}2300$.

To economists explicit cost is not only the cost. Economic cost includes explicit and implicit cost. An implicit cost represents the value of resources used in production for which no actual (monetary) payment is made. It is a cost incurred as a result of a firm's using resources that it owns or that the owners of the firm contribute to it.

Accounting Profit and Economic Profit

Accounting profit is the difference between total revenue and total cost, where total cost equals explicit costs [see Exhibit 1(a)]:

$$\text{Accounting profit} = \text{Total revenue} - \text{Total cost (Explicit costs)}$$

Economic profit is the difference between total revenue and total cost, where total cost equals the sum of explicit and implicit costs [see Exhibit 1(b)]:

$$\text{Economic profit} = \text{Total revenue} - \text{Total cost (Explicit costs + Implicit costs)}$$

To illustrate the difference between explicit and implicit costs, suppose a person has \$100,000 in the bank, earning an interest rate of 5 percent a year. This amounts to \$5,000 in interest a year. Now suppose the person takes the \$100,000 out of the bank to start a business. The \$5,000 in lost interest is included in the implicit costs of owning and operating the firm. Suppose the person does not use her \$100,000 in the bank to start a business but leaves it in the bank and instead takes out a \$100,000 loan at an interest rate of 5 percent. Then the interest she has to pay on the loan—\$5,000 a year—certainly would be an explicit cost and would take away from her overall profit. It just makes sense, then, to count the \$5,000 interest that the owner doesn't earn if she uses her own \$100,000 to start the business (instead of taking out a loan) as a cost, albeit implicit.

Zero Economic Profit or Normal Profit

In economics, a firm that makes a zero economic profit is said to be earning a normal profit.

$$\text{Normal profit} = \text{Zero economic profit}$$

Zero economic profit, the level of profit necessary to keep resources employed in a firm. A firm that earns normal profit is earning revenue equal to its total costs (explicit plus implicit costs).

$$\text{I.e., Normal profit (=Zero economic profit)} = \text{TR} = \text{TC}$$

However, the owner of a firm should not be worried about making zero economic profit for the year just ending. A zero economic profit is not that bad as it sounds—means that the owner has generated total revenue sufficient to cover total cost—that is, both explicit and implicit costs. If, for example, the owner's implicit cost is a (forfeited) \$100,000 salary working for someone else, then earning a zero economic profit means that he has done as well as he could have done in his next-best (alternative) line of employment.

When we realize that zero economic profit (or normal profit) means doing as well as could have been done, we understand that it isn't bad to make zero economic profit. Zero accounting profit, however, is altogether different: It implies that some part of total cost has not been covered by total revenue.

PRODUCTION

Production is a transformation of resources or inputs into goods and services. It takes certain ingredients to make a cake: sugar, flour, and so on. Similarly, it takes certain resources, or inputs, to produce a computer.

There are two types of inputs in the production process: **fixed and variable**. A **fixed input** is an input whose quantity cannot be changed as output changes. For example, rented factory is a fixed input. A **variable input** is an input whose quantity can be changed as output changes. Labor might be a variable input. As a producer produces more of a good he they might hire more employees; he produces fewer he might lay off some.

If any of the inputs of a firm are fixed inputs, then it is said to be producing in the **short run**. In other words, the **short run** is a period during which some inputs are fixed. If none of the inputs of a firm is a fixed input—if all inputs are variable—then the firm is said to be producing in the **long run**. In other words, the **long run** is a period during which all inputs can be varied. (No inputs are fixed).

Production in the Short Run

Suppose two inputs (or resources)—labor (L) and capital (C)—are used to produce some good. Suppose one of those inputs—capital—is fixed. Obviously, because an input is fixed, the firm is producing in the short run.

In Exhibit 2, column 1 shows the units of the fixed input, capital (fixed at 1 unit). Column 2 shows different units of the variable input, labor. Notice that we go from 0 (no workers) through 10 units (10 workers). Column 3 shows the quantities of output produced with 1 unit of capital and different amounts of labor. The quantity of output is sometimes referred to as the total product, or TP.

EXHIBIT 2

**Production in the Short Run
and the Law of Diminishing
Marginal Returns**

In the short run, as additional units of a variable input are added to a fixed input, the marginal physical product of the variable input may increase at first. Eventually, the marginal physical product of the variable input decreases. The point at which marginal physical product decreases is the point at which diminishing marginal returns have set in.

(1) Capital (C), fixed input (units)	(2) Labor (L), variable input (number of workers)	(3) Quantity of Output, Q (units)	(4) Marginal Physical Product (MPP) of Labor, MPP of labor = $\Delta Q / \Delta L$, $\Delta(3) / \Delta(2)$ (units)
1	0	0	
1	1	18	18
1	2	37	19
1	3	57	20
1	4	76	19
1	5	94	18
1	6	111	17
1	7	127	16
1	8	137	10
1	9	133	-4
1	10	125	-8

For example, 1 unit of capital and 0 units of labor produce 0 output; 1 unit of capital and 1 unit of labor produce 18 units of output; 1 unit of capital and 2 units of labor produce 37 units of output; 1 unit of capital and 3 units of labor produce 57 units of output; and so on.

Column 4 shows the marginal physical product In short marginal product of the variable input. The marginal physical product (MPP) of a variable input is equal to the change in output that results from changing the variable input by one unit while holding all other inputs fixed. Because, in our example, the variable input is labor, we are talking about the MPP of labor. Specifically, the **MPP** or **MP** of labor is equal to the change in output Q that results from changing labor L by one unit while holding all other inputs fixed:

$$\text{MPP of labor} = \Delta Q / \Delta L$$

Notice that the marginal physical product of labor first rises (from 18 to 19 to 20), then falls (from 20 to 19 to 18 to 17 to 16 to 10), and then becomes negative (-4 and -8). When the MPP is rising, we say that there is increasing MPP; when it is falling, there is diminishing MPP; and when it is negative, there is negative MPP.

As we see from the exhibit the MPP begins to decline with the addition of the fourth worker. The point at which the MPP of labor first declines is the point at which **diminishing marginal returns** are said to have set in. The **Law of Diminishing marginal returns** (or the law of diminishing marginal product) are common in production. **The law of diminishing marginal returns** states that as ever larger amounts of a variable input

are combined with fixed inputs, eventually the marginal physical product of the variable input will decline.

Why does the MPP of the variable input eventually decline? To answer this question, think of adding agricultural workers (variable input) to 10 acres of land (fixed input). In the early stages of adding labor to the land, perhaps the MPP rises or remains constant. But eventually, as we continue to add more workers to the land, it becomes overcrowded with workers. Workers are stepping around each other, stepping on the crops, and so on. Because of these problems, output growth begins to slow.

It may seem strange that the firm in Exhibit 2 would ever hire beyond the third worker. After all, the MPP of labor is at its highest (20) with the third worker. Why hire the fourth worker if the MPP of labor is going to fall to 19? The firm may hire the fourth worker because the worker adds output. It would be one thing if the quantity of output were 57 units with three workers and fell to 55 units with the addition of the fourth worker, but this isn't the case. With the addition of the fourth worker, output rises from 57 units to 76 units. The firm has to ask and answer two questions: (1) What can the additional 19 units of output be sold for? (2) What does it cost to hire the fourth worker? Suppose the additional 19 units can be sold for \$100, and it costs the firm \$70 to hire the fourth worker. In that case, hiring the fourth worker makes sense.

Marginal Physical Product and Marginal Cost

A firm's costs are tied to its production. Specifically, the marginal cost (MC) of producing a good is a reflection of the MPP of the variable input. Certainly, a cost is incurred whenever a fixed input or variable input is employed in the production process. The costs associated with fixed inputs are called **fixed costs**. The costs associated with variable inputs are called **variable costs**.

Because the quantity of a fixed input does not change as output changes, neither do fixed costs change with output. Payments for such things as fire insurance (the same amount every month), liability insurance, and the rental of a factory and machinery are usually considered **fixed costs**.

The quantity of a variable input changes with output, so do variable costs. For example, it takes labor, wood, and glue to produce wooden bookshelves. The quantity of all these inputs (labor, wood, and glue) changes as the number of wooden bookshelves produced changes.

The sum of fixed costs and variable costs is **total cost (TC)**. If total fixed costs (TFC) are \$100 and total variable costs (TVC) are \$300, then total cost (TC) is \$400:

$$TC = TFC + TVC = \$100 + \$300 = \$400$$

Marginal cost (MC) is the change in total cost TC that results from a change in quantity of output Q:

$$MC = \frac{\Delta TC}{\Delta Q}$$

Relationship Between MPP and MC

MC is the additional cost of an additional unit of the variable input. MPP is the additional output produced by an additional input employed. In Exhibit 3, the variable input is labor, so $MC = \frac{W}{MPP_L}$ where MC = marginal cost, W = wage, and MPP_L = marginal physical product of labor.

EXHIBIT 3

Marginal Physical Product and Marginal Cost

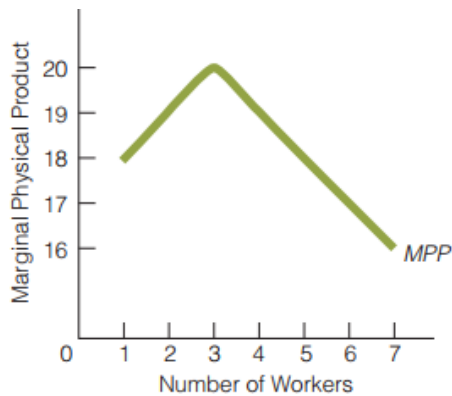
(a) The marginal-physical-product-of-labor

curve. The curve is derived by plotting the data from columns 2 and 4 in the table.

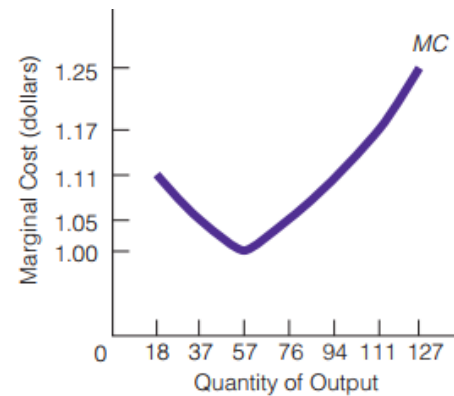
(b) The marginal cost curve. The curve is derived by plotting the data from columns

3 and 8 in the table. Notice that as the MPP curve rises, the MC curve falls; and as the MPP curve falls, the MC curve rises.

(1) Capital (C), fixed input (units)	(2) Labor (L), variable input (workers)	(3) Quantity of Output, Q (units)	(4) Marginal Physical Product (MPP) of Labor, MPP of labor = $\Delta Q / \Delta L =$ $\Delta(3) / \Delta(2)$ (units)	(5) Total Fixed Cost (TFC)	(6) Total Variable Cost (TVC)	(7) Total Cost (TC), (5)+(6)	(8) Marginal Cost (MC), $\Delta TC / \Delta Q$, $\Delta(7) / \Delta(3)$
1	0	0		\$40	\$0	\$40	
1	1	18	18	40	20	60	\$1.11
1	2	37	19	40	40	80	\$1.05
1	3	57	20	40	60	100	\$1.00
1	4	76	19	40	80	120	\$1.05
1	5	94	18	40	100	140	\$1.11
1	6	111	17	40	120	160	\$1.17
1	7	127	16	40	140	180	\$1.25



(a)



(b)

The following table reproduces column 4 from Exhibit 3, notes the wage, and uses the equation $MC = W/MPP$ to compute MC :

<i>MPP</i>	Variable Cost (<i>W</i>)	$W/MPP = MC$
18 units	\$20	$\$20/18 = \1.11
19	20	$20/19 = 1.05$
20	20	$20/20 = 1.00$
19	20	$20/19 = 1.05$
18	20	$20/18 = 1.11$
17	20	$20/17 = 1.17$
16	20	$20/16 = 1.25$

As we compare column 4 and column 8 of the table in exhibit and also figure (a) and (b) we observe that as MPP rises MC falls, as MPP falls, MC rises.

$$\frac{W}{MP \uparrow} = MC \downarrow$$

$$\frac{W}{MP \downarrow} = MC \uparrow$$

Average Productivity

Average productivity of an input is total output produced by an input divided by total amount of that inputs.

Average productivity = total output/total amount of an input, for example, average productivity of labor:

$$AP \text{ of labor} = \frac{Q}{L}$$

COSTS OF PRODUCTION: TOTAL, AVERAGE, MARGINAL

In exhibit 4, column 1 of Exhibit 4 shows the various quantities of output, ranging from 0 units to 10 units. Column 2 shows the total fixed costs of production (TFC), set at \$100. TFC does not change as output, Q changes, the TFC curve in the exhibit is a horizontal line at \$100.

In column 3, average fixed cost (AFC), which is total fixed cost divided by quantity of output:

$$AFC = \frac{TFC}{Q}$$

The AFC curve continually declines as output increases.

EXHIBIT 4

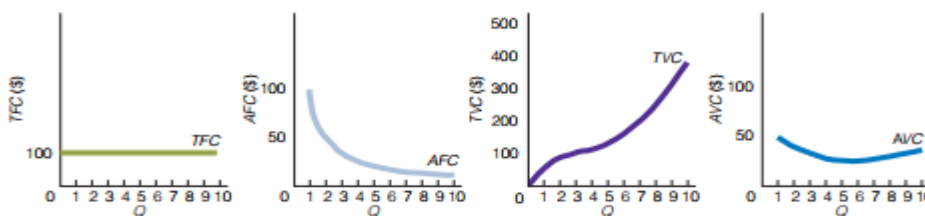
Total, Average, and Marginal Costs

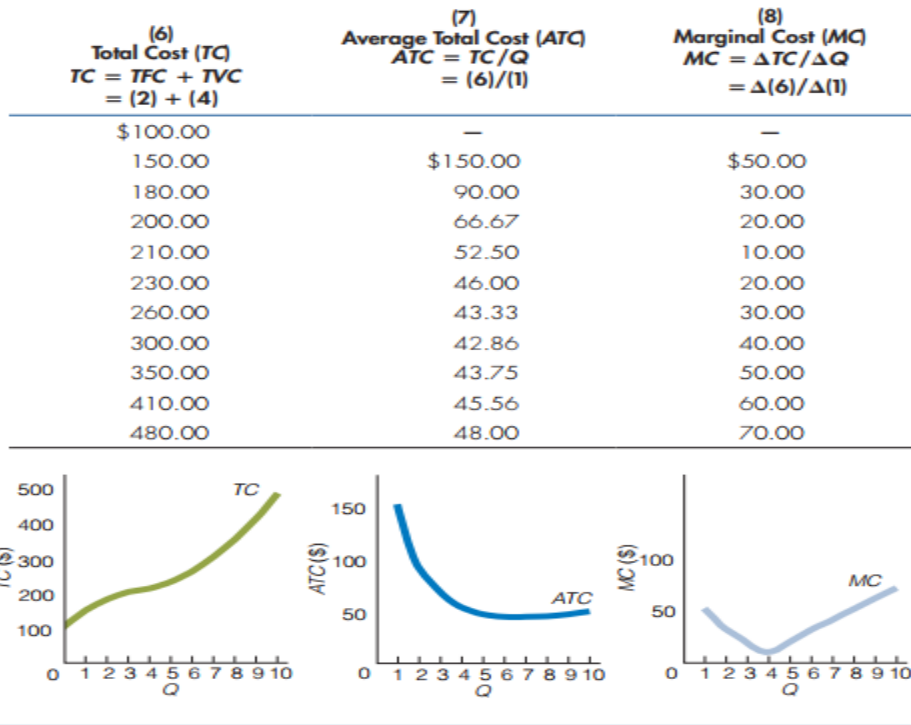
TFC equals \$100 (column 2 on the table) and TVC is as noted in column 4. From the

data, we calculate AFC, AVC, TC, ATC, and MC. The curves associated with TFC, AFC, TVC, AVC, TC, ATC, and MC are

shown in diagrams at the bottom of the corresponding columns. (Note: Scale is not the same for all diagrams.)

(1) Quantity of Output, Q (units)	(2) Total Fixed Cost (TFC)	(3) Average Fixed Cost (AFC) $AFC = TFC / Q$ $= (2)/(1)$	(4) Total Variable Cost (TVC)	(5) Average Variable Cost (AVC) $AVC = TVC / Q$ $= (4)/(1)$
0	\$100	—	\$0	—
1	100	\$100.00	50	\$50.00
2	100	50.00	80	40.00
3	100	33.33	100	33.33
4	100	25.00	110	27.50
5	100	20.00	130	26.00
6	100	16.67	160	26.67
7	100	14.28	200	28.57
8	100	12.50	250	31.25
9	100	11.11	310	34.44
10	100	10.00	380	38.00





Column 4 shows total variable cost (TVC). TVC increases with the increase in output. The TVC curve is upward sloping.

Column 5 shows average variable cost (AVC), which is total variable cost divided by quantity of output:

$$AVC = \frac{TVC}{Q}$$

Column 6 shows total cost (TC). Total cost is the sum of total variable cost and total fixed cost. The TC curve does not start at zero because, even when output is zero, there are some fixed costs. In this example, total fixed cost (TFC) at zero output is \$100, so the total cost (TC) curve must start at \$100 instead of at \$0.

Column 7 shows average total cost (ATC), which is total cost divided by quantity of output:

$$ATC = \frac{TC}{Q}$$

Average total cost is sometimes called unit cost. Alternatively, we can say that ATC equals the sum of AFC and AVC:

$$ATC = AFC + AVC$$

Or,

$$\frac{TC}{Q} = \frac{TFC}{Q} + \frac{TVC}{Q}$$

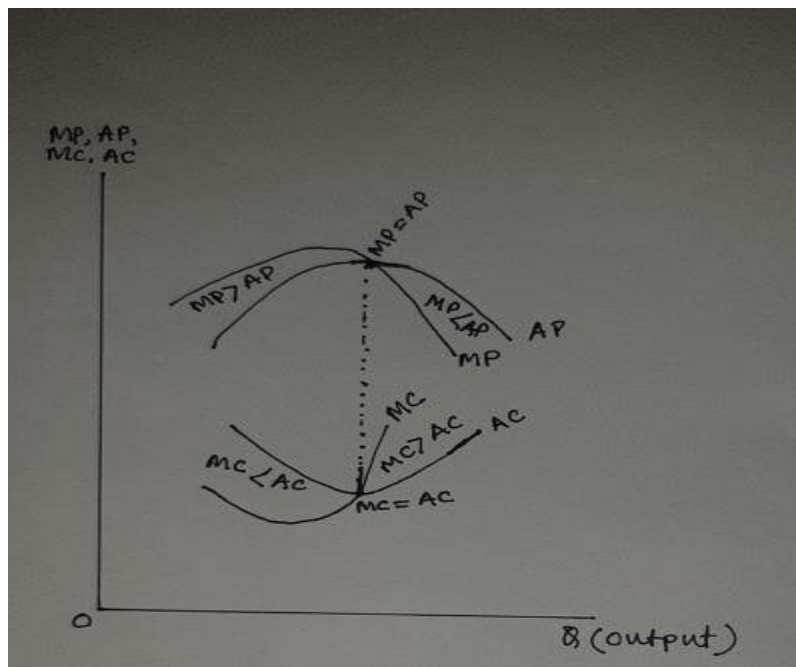
Column 8 shows marginal cost (MC), which is the change in total cost divided by the change in quantity of output:

$$MC = \frac{\Delta TC}{\Delta Q}$$

The MC curve has a declining portion and a rising portion. When MC is declining, MPP is rising. When MC is rising, the MPP of the variable input is falling. Obviously, the low point on the MC curve is when diminishing marginal returns set in.

Sunk cost: A cost incurred in the past that cannot be changed by current decisions and therefore cannot be recovered.

Relationship between MP and AP; MC and AC and MP/AP and MC/AC



As long as MP is rising AP is also rising and $MP > AP$. At the highest point of AP, MP intersects AP. At the intersection point $MP = AP$. Beyond that as MP falls AP also falls and $MP < AP$.

As long MC is falling AC is also falling and $MC < AC$. At the lowest point of AC, MC intersects AC. At this point $MC = AC$. Beyond that as MC rises and AC also rises and $MC > AC$.

When MP and AP are rising, MC and AC are falling. When $MP = AP$, $MC = AC$. Beyond that if MP and AP fall, MC and AC rise.

PRODUCTION AND COSTS IN THE LONG RUN

There are no fixed inputs and no fixed costs in the long run. Consequently, the firm has greater flexibility in the long run than in the short run.

Long-Run Average Total Cost Curve

In the short run, because there are fixed costs and variable costs, total cost is the sum of the two. But in the long run, there are no fixed costs, so variable costs are total costs. This section focuses on (1) the long-run average total cost (LRATC) curve and (2) what it looks like.

When all inputs are variable, the firm manager must decide what the situation of the firm should be in the upcoming short-run period. For example, he might need to determine the size of the plant: small, medium, or large. Once this decision is made, the firm is locked into a specific plant size for the short run. Associated with each of the three different plant sizes is a short-run average total cost (SRATC) curve, as illustrated in Exhibit 8 (a).

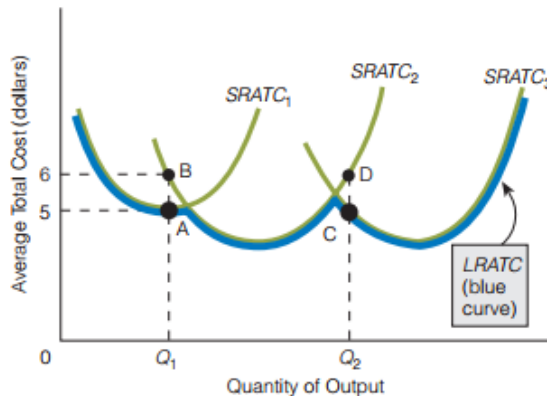
EXHIBIT 8

Long-Run Average Total Cost Curve (LRATC)

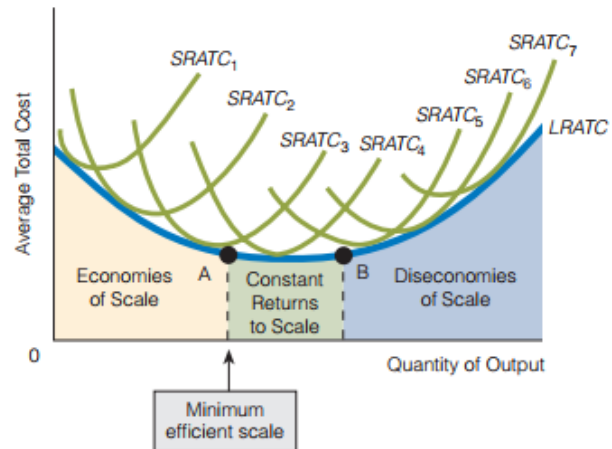
(a) There are three short-run average total cost curves, for three different plant sizes. If these are the

only plant sizes, then the long-run average total cost curve is the heavily shaded, blue scalloped curve. (b) The long-run average total cost curve is the heavily shaded, blue

smooth curve. The LRATC curve in (b) is not scalloped because it is assumed that there are so many plant sizes that the LRATC curve touches each SRATC curve at only one point.



(a)



(b)

Suppose the manager of the firm wants to produce output level Q_1 . Then, obviously, he will choose the plant size represented by $SRATC_1$. This size gives a lower unit cost of producing Q_1 than the plant size represented by $SRATC_2$, a size that has a higher unit cost of producing Q_1 (\$6 as opposed to \$5). However, if the manager chooses to produce Q_2 , he will choose the plant size represented by $SRATC_3$, because the unit cost of producing Q_2 is lower with that plant size than it is with the plant size represented by $SRATC_2$.

If we were to ask the same question for every (possible) output level, we would derive the long-run average total cost (LRATC) curve. **The LRATC curve** shows the lowest unit cost at which the firm can produce any given level of output. In Exhibit 8(a), the LRATC curve consists of the portions of the three SRATC curves that are tangential to the LRATC curve.

Exhibit 8(b) shows a host of SRATC curves and one LRATC curve. The LRATC curve is smooth in part (b) because we assume that there are many plant sizes in addition to the three represented in (a). In other words, although they have not been drawn, short-run average total cost curves representing different plant sizes exist in (b) between $SRATC_1$ and $SRATC_2$, between $SRATC_2$ and $SRATC_3$, and so on. In this case, the LRATC curve is smooth and touches each SRATC curve at one point.

Suppose two inputs—labor and capital—are used together to produce a good. If inputs are increased by some percentage (say, 100 percent) and if output increases by a greater percentage (more than 100 percent), then unit costs fall and economies of scale are said to exist. For example, suppose that good X is made with two inputs, Y and Z, and it takes 20 Y and 10 Z to produce 5 units of X. Suppose also that the cost of each unit of input Y is \$1 and the cost of each unit of input Z is \$1. Thus, a total cost of \$30 is required to produce 5 units of X. The unit cost (average total cost) of good X is \$6 ($ATC = TC/Q = 30/5 = \6). Now, consider a doubling of inputs Y and Z, to 40 Y and 20 Z, and a more than doubling in output, say, to 15 units of X. Then a total cost of \$60 is required to produce 15 units of X, and the unit cost (average total cost) of good X is \$4.

An increase in inputs can have two other results. If inputs are increased by some percentage and output increases by an equal percentage, then unit costs remain constant and constant returns to scale are said to exist. If inputs are increased by some percentage and output increases by a smaller percentage, then unit costs rise and diseconomies of scale are said to exist.

The three conditions can easily be seen in the LRATC curve in Exhibit 8(b): if economies of scale are present, the LRATC curve is falling; if constant returns to scale are present, the curve is flat; if diseconomies of scale are present, the curve is rising. In sum,

Economies of scale → LRATC is falling
 Constant returns to scale → LRATC is constant
 Diseconomies of scale → LRATC is rising

If, in the production of a good, economies of scale give way to constant returns to scale or to diseconomies of scale, as in Exhibit 8(b), the point at which this transition occurs is referred to as the minimum efficient scale. **The minimum efficient scale** is the lowest output level at which average total costs are minimized. Point A represents the minimum efficient scale in Exhibit 8(b).

The significance of the minimum efficient scale of output can be seen by looking at the long-run average total cost curve between points A and B in Exhibit 8(b). Between these points, there are constant returns to scale: the average total cost is the same over the various output levels between the two points. This means that larger firms (firms producing greater output levels) within the range from point A to point B do not have a cost advantage over smaller firms that operate at the minimum efficient scale.

Economies of scale, diseconomies of scale, and constant returns to scale are relevant only in the long run.

Why does economies of scale occur?

Up to a certain point, long-run unit costs of production fall as a firm grows, for two main reasons: (1) Growing firms offer greater opportunities for employees to specialize. Individual workers can become highly proficient at narrowly defined tasks, often producing more output at lower unit costs. (2) Growing firms (especially, large growing firms) can take advantage of highly efficient mass production techniques and equipment that ordinarily require large setup costs and thus are economical only if they can be spread over a large number of units. For example, assembly-line techniques are usually relatively cheap when millions of units of a good are produced but are expensive when only a few thousand units are produced.

Why Diseconomies of Scale?

Diseconomies of scale usually arise at the point where a firm's size causes coordination, communication, and monitoring problems. In very large firms, managers often find it difficult to coordinate work activities, communicate their directives to the right persons in a timely way, and monitor personnel effectively. The business operation simply gets too big. There is, of course, a monetary incentive not to pass the point of operation at which diseconomies of scale exist, and firms usually find ways to do so, including reorganizing, dividing operations, hiring new managers, and more.

SHIFTS IN COST CURVES

Following factors shift cost curves:

Taxes

Consider a tax on each unit of a good produced. Suppose, for example, that a company has to pay a tax of \$3 for each unit of X it produces. What effects will this tax have on the firm's cost curves? The tax won't affect the firm's fixed costs, because the tax is paid only when output is produced—and fixed costs are present even if output is zero. (If the tax were a lump-sum tax—that is, if the company pays a lump sum no matter how many units of X it produces—then the tax would affect fixed costs.) So, given that the tax is not a lump-sum tax, we conclude that the tax does not affect fixed costs and therefore cannot affect average fixed cost.

The tax, however, will affect variable costs. As a consequence of the tax, the firm has to pay more for each unit of X it produces. Because variable costs rise, so does total cost. This relationship means that average variable cost and average total cost rise and that the representative cost curves shift upward. Finally, because marginal cost is the change

in total cost divided by the change in output, marginal cost rises and the marginal cost curve shifts upward.

Input Prices

A rise or fall in variable input prices causes a corresponding change in the firm's average total, average variable, and marginal cost curves. For example, if the price of steel rises, the variable costs of building skyscrapers rise and so must average variable cost, average total cost, and marginal cost. The cost curves shift upward. If the price of steel falls, the opposite effects occur.

Technology

Technological changes often bring either (1) the capability of using fewer inputs to produce a good (e.g., the introduction of the personal computer reduced the hours necessary to key and edit a manuscript) or (2) lower input prices (e.g., technological improvements in transistors led to price reductions in the transistor components of calculators). In either case, technological changes of this variety lower variable costs and consequently lower average variable cost, average total cost, and marginal cost. The cost curves shift downward.

Sample questions for quiz

Define a business Firm

A business Firm is an entity that employs factors of production (resources) to produce goods and services to be sold to consumers, other firms, or the government.

What is an explicit cost?

An explicit cost is a cost that is incurred when an actual (monetary) payment is made.

What is an implicit cost?

An implicit cost represents the value of resources used in production for which no actual (monetary) payment is made. It is a cost incurred as a result of a firm's using resources that it owns or that the owners of the firm contribute to it.

What is an Economic cost?

Economic cost = explicit + Implicit cost.

What is an Economic profit?

$$\text{Economic profit} = \text{Total revenue} - \text{Total cost (Explicit costs + Implicit costs)}$$

What is production?

Production is a transformation of resources or inputs into goods and services. It takes certain ingredients to make a cake: sugar, flour, and so on.

Define fixed input and variable input

A fixed input is an input whose quantity cannot be changed as output changes. For example, rented factory is a fixed input. A variable input is an input whose quantity can be changed as output changes. Labor might be a variable input.

What do you mean by the law of diminishing marginal returns?

The law of diminishing marginal returns states that as ever larger amounts of a variable input are combined with fixed inputs, eventually the marginal physical product of the variable input will decline.

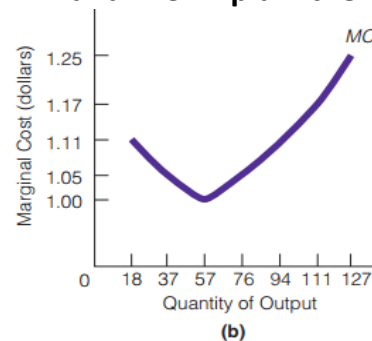
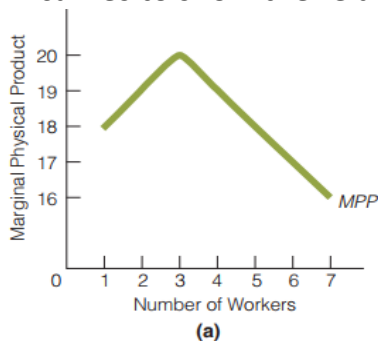
Define zero or normal profit

A zero economic profit or normal profit means that the owner has generated total revenue sufficient to cover total cost—that is, both explicit and implicit costs.

Define fixed and variable costs.

The costs associated with fixed inputs are called **fixed costs**. The costs associated with variable inputs are called **variable costs**.

Draw curves to show the relationship between MP and MC. Explain the relationship.



$$\frac{W}{MP} \uparrow = MC \downarrow$$

$$\frac{W}{MP} \downarrow = MC \uparrow$$

As MP rises, MC falls and vice versa.

Why are short-run and long-run average costs (AC) are U-shaped?

Short-run average cost curve is U-shaped because of

(a) Average fixed cost is decreasing and (b) the law of diminishing marginal returns.

Long-run average cost curve is U-shaped because of

(a) Economies of scale (b) constant returns to scale and (c) diseconomies of scale.

What do you mean by economies of scale?

If inputs are increased by some percentage but output increases by a greater percentage then unit costs fall and economies of scale are said to exist.

What do you mean by diseconomies of scale?

If inputs are increased by some percentage and output increases by a smaller percentage, then unit costs rise and diseconomies of scale are said to exist.

Why do economies of scale occur?

Up to a certain point, long-run unit costs of production fall as a firm grows, for two main reasons: (1) Growing firms offer greater opportunities for employees to specialize, often producing more output at lower unit costs. (2) Growing firms (especially, large growing firms) can take advantage of highly efficient mass production techniques and equipment.

Why do diseconomies of scale occur?

Diseconomies of scale usually arise at the point where a firm's size causes coordination, communication, and monitoring problems. In very large firms, managers often find it difficult to coordinate work activities, communicate their directives to the right persons in a timely way, and monitor personnel effectively.

Why do cost curves shift?

Cost curves shift because of (a) Taxes (b) input prices and (c) technology.